

# Software Engineering Task 1 2025

William Lam

## Requirements Definition

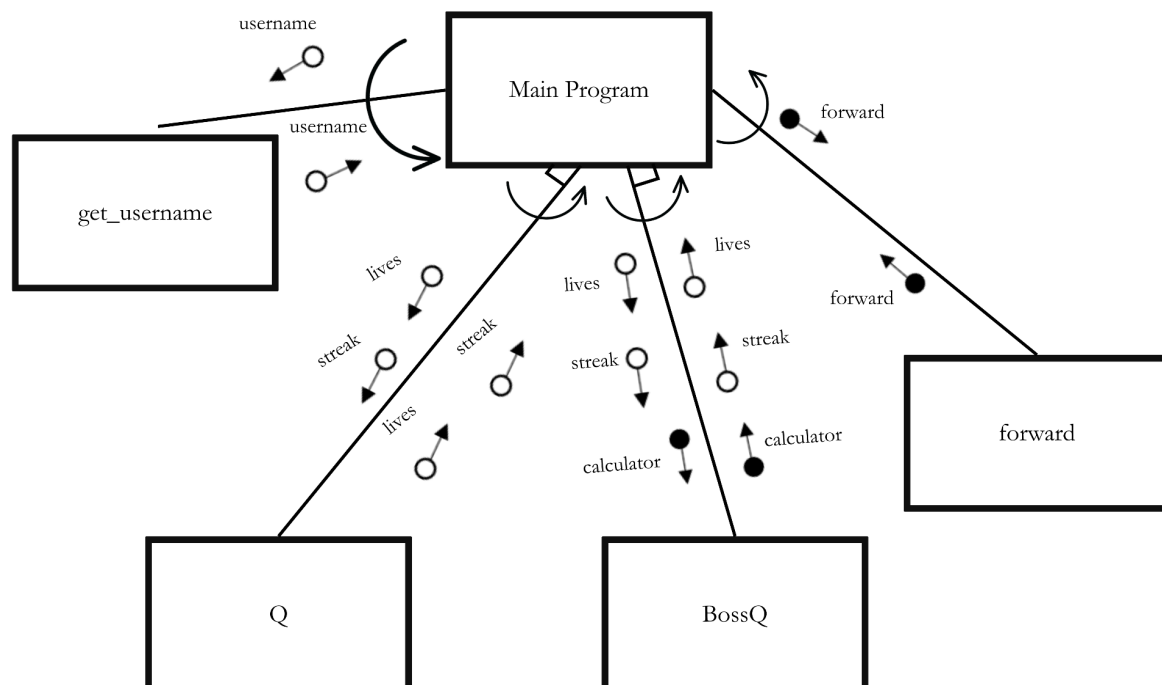
- Create a game for primary students that is both fun and educational
- A story based game, where the users complete puzzles to advance in the story
  - Antagonist, asks user to do math problems
- Minimum of 5 questions
- Questions must be easy to interpret and answer
- Point scoring system, different outcomes for wrong answers
- Have the item: Calculator - item which allows the user to get one question free
  - Gotten after getting 5 in a row
  - Maximum of one

## User Specifications

- Features and Functions
  - 10 questions in total
    - 7 'normal'
    - 3 'hard' (AKA bossfight questions)
  - If the user has a current streak of 2 after the 7 normal questions, they will receive an extra life
  - If above 5, calculator will be given
  - All answers will be either integers or floats
- How it works
  - Users will be asked a series of questions
  - Goal is to correctly answer all 10 to defeat the 'Math Inquisitor'
  - 5 lives
  - A streak counter is used, reset every time a question is answered incorrectly
  - Calculator - item that is achieved after getting a streak of 5, allows a free pass on an boss fight question
  - Data is saved to sequential file and displayed at the beginning of the program
- Look and Feel
  - Colours will be default
  - No buttons
  - Will rely on keyboard interaction
- Speed and Performance

- Should operate instantly
- Program can be duplicated so that it can run by different users at the same time
- Security
  - It will not be collecting any personal data, so there is no real need for security
- Compatibility
  - It will run on personal computers via python

## Structure Chart



## Logbook

| Date       | What was completed                              | Challenges                                                        | Areas for Improvement           |
|------------|-------------------------------------------------|-------------------------------------------------------------------|---------------------------------|
| 31/03/2025 | Folio was created<br>Idea for the game was made | Thinking of an idea that was feasible in the time was challenging | Working in a fast enough manner |

|            |                                                                                               |                                                                                                              |                                                                       |
|------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 01/04/2025 | Requirements definition and user specifications were created                                  | Making sure the definitions would fit the game                                                               | User specifications could be more detailed                            |
| 02/04/2025 | User specifications were completed<br>The first question for the game was created and tested. | Making the code of the question in a way that would allow it to be easily replicated for the other questions | Making the code modular.                                              |
| 03/04/2025 | Pseudocode was finished and began turning it into python code                                 | Due to a lot of code being repeated, they were left out to be finished during python coding                  | Making sure all the unique points of code were made in the pseudocode |
| 04/04/2025 | Continued turning pseudocode into python code                                                 | A lot of missing things had to be added in the python code such as white space                               | Working at a faster pace                                              |
| 06/04/2025 | Python code was completed                                                                     | Testing the code took time                                                                                   | Working through the code, fixing errors faster                        |
| 07/04/2025 | Structure chart was made                                                                      | Instead of modelling the code on the structure chart, the structure chart was modelled on the python code    | Next time, do the structure chart before doing the python code        |
| 08/04/2025 | Desk checks were started                                                                      | Desk checking was extremely time consuming                                                                   | Work faster on desk checks                                            |
| 09/04/2025 | Desk checks were finished<br>Assignment was submitted                                         | Desk checking was still time consuming                                                                       | NIL                                                                   |

## Pseudocode

```

BEGIN get_username(username)
    PRINT("Enter Username")
    Get username
    RETURN username
END get_username()

BEGIN forward(forward)
    forward = FALSE
    WHILE forward != ""
        PRINT("Press Enter to continue")
        Get forward
    forward = TRUE
    RETURN forward
END forward()

```

```

BEGIN Q1(lives, streak)
    correct_answer = 30
    WHILE lives > 0
        PRINT("Question One")
        PRINT("5 + 25")
        Get answer
        IF answer != correct_answer THEN
            streak = 0
            lives = lives - 1
            PRINT("Incorrect Answer! Streak Lost!")
            IF lives == 1 THEN
                PRINT("You have 1 life remaining")
            ELSE
                PRINT("You have lost one life. You have", lives, "lives remaining")
            ENDIF
        ELSE
            PRINT("Correct!")
            PRINT("You have passed Question One")
            streak = streak + 1
            RETURN lives, streak
        ENDIF
    ENDWHILE
    IF lives <= 0
        PRINT "Game Over"
        RETURN (lives, streak)
    END Q1(lives, streak)

```

```

BEGIN bossQ1(lives, streak, calculator)
    correct_answer = 7
    WHILE lives > 0:
        PRINT("Question 1")
        PRINT("2(1+3) - 7/(4+3) + (1+3) x 0")
        choice = ""
        WHILE choice != 'A' or 'C':
            PRINT('Input A to answer')
            PRINT('Input C to use calculator')
            Get choice
            choice = choice.upper()

        IF choice == 'C' and calculator == True THEN
            calculator = False
            streak = streak + 1
            PRINT("You have passed using a CALCULATOR. Lucky!")
            answer = correct_answer
            return(lives, streak, calculator)
        ELSE IF choice == 'C' and calculator == False:
            PRINT("You do not have a calculator")
            choice = ""
        ELSE IF choice == 'A' THEN
            PRINT("Enter your answer")
            Get answer
            IF answer != correct_answer THEN
                streak = 0
                lives = lives - 1
                PRINT("Incorrect Answer! Streak Lost!")
                IF lives == 1 THEN
                    PRINT("You have 1 life remaining.")
                ELSE
                    PRINT("You have lost one life. You have", lives, "lives remaining.")
                ENDIF
            ELSE
                PRINT("Correct!")
                PRINT("You have passed Question One.")
                streak = streak + 1
                return (lives, streak, calculator)
            ENDIF
        ENDIF
    ENDWHILE
    IF lives <= 0 THEN
        PRINT("Game Over.")
        return (lives, streak, calculator)
    END bossQ1(lives, streak, calculator)

```

```

BEGIN main

    begin = 0

```

```

username = "
username_confirmed = 'N'
lives = 5
streak = 0

PRINT("Welcome to Math Quest!")
PRINT("Answer a series of questions to save the land from the treacherous Math Inquisitor!")
PRINT("Good luck brave soul...")
PRINT()
WHILE begin != "
    PRINT("Press ENTER to start")
    Get begin
    IF begin != " THEN
        PRINT()
        PRINT("Waiting...")
        PRINT()
    ENDIF
ENDWHILE

PRINT()
PRINT("Loading...")
PRINT()

f = open("statistics.txt", "a")
f.close()
PRINT("Showing previous scores. If nil, there are no previous scores. Statistics will only be recorded if the entire run is finished.")
f = open("statistics.txt", "r")
FOR x in f:
    PRINT(x)
ENDFOR
f.close()
PRINT()

WHILE username_confirmed != "Y"
    username = get_username()
    PRINT("Confirm username as: ", username, "? (Y/N)")
    Get username_confirmed.upper()
ENDWHILE

PRINT("Username is confirmed as ", username)

PRINT("Time starts now")
start= time.time()

WHILE lives > 0 and questions_finished == False

    lives, streak = Q1(lives, streak)
    PRINT "Stats!"
    PRINT "Your current answer streak is " + streak
    PRINT "Lives remaining: " + lives

    questions_finished = TRUE

ENDWHILE

IF lives <= 0 THEN
    EXIT
ENDIF

IF streak >= 2 THEN
    lives = lives + 1
    PRINT "Since you have a streak of 2, you got an extra life!"
    PRINT "You now have " + lives + " lives"
ENDIF

IF streak >= 5 THEN
    calculator = TRUE
    PRINT "Since you have a streak of 5, you have received a CALCULATOR. Use it in the bossfight to immediately answer any question!"
ELSE
    calculator = FALSE
ENDIF

PRINT "Loading BOSS FIGHT"
PRINT 'Here lies the Math Inquisitor, the final boss. Beware of his deadly questions..'

questions_finished = FALSE

WHILE lives > 0 and questions_finished == FALSE

    lives, streak, calculator = bossQ1(lives, streak, calculator)
    PRINT "Stats!"
    PRINT "Your current answer streak is " + streak

```

```

        PRINT "Lives remaining: " + lives
        continueon = forward(forward)

        questions_finished = TRUE
    ENDWHILE

    IF lives <= 0 THEN
        EXIT
    ENDIF

    end = time.time()

    PRINT "You have defeated the Math Inquisitor! You lie the King of the land."
    PRINT "Your time was " + format((end_time-start_time)/60, ",.2f") + " minutes."
    PRINT "Your statistics have been recorded"
    PRINT "GAME COMPLETED"

    statistics_array = array [username,_streak,_lives,_end_time]
    file = openWrite("statistics.txt")
    file.writeLine(str(statistics_array))
    file.close
    file = openWrite("statistics.txt")

END main

```

## Desk Check

get\_username()

Test Data

|          | username | Expected Output |
|----------|----------|-----------------|
| Data Set | W1lliam  |                 |

Desk Check

| Line No. | username | Output           |
|----------|----------|------------------|
| 1        |          | "Enter Username" |
| 2        | W1lliam  |                  |
| 3        |          |                  |

forward()

Test Data

|  | forward | Expected Output |
|--|---------|-----------------|
|--|---------|-----------------|

|          |     |  |
|----------|-----|--|
| Data Set | “ “ |  |
|----------|-----|--|

Desk Check

| Line No. | forward | Expected Output           |
|----------|---------|---------------------------|
| 1        | FALSE   |                           |
| 2        |         |                           |
| 3        |         | “Press Enter to continue” |
| 4        | “ “     |                           |
| 5        | TRUE    |                           |
| 6        |         |                           |

Q1()

Test Data

|            | correct_answer | lives | answer | streak | Expected Output                |
|------------|----------------|-------|--------|--------|--------------------------------|
| Data Set 1 | 30             | 5     | 30     | 0      | “You have passed Question One” |
| Data Set 2 | 30             | 5     | 29     | 0      | “Game Over”                    |

Desk Check

| Data Set 1 |                |       |        |        |           |                         |            |                 |
|------------|----------------|-------|--------|--------|-----------|-------------------------|------------|-----------------|
| Line No.   | correct_answer | lives | answer | streak | lives > 0 | answer = correct_answer | lives <= 0 | Expected Output |
| 1          | 30             | 5     |        | 0      |           |                         |            |                 |
| 2          |                |       |        |        | TRUE      |                         |            |                 |

|    |  |  |    |   |  |      |  |                                              |
|----|--|--|----|---|--|------|--|----------------------------------------------|
| 3  |  |  |    |   |  |      |  | "Questi<br>on One"                           |
| 4  |  |  |    |   |  |      |  | "5+25"                                       |
| 5  |  |  | 30 |   |  |      |  |                                              |
| 6  |  |  |    |   |  | TRUE |  |                                              |
| 15 |  |  |    |   |  |      |  |                                              |
| 16 |  |  |    |   |  |      |  | "Correct<br>"                                |
| 17 |  |  |    |   |  |      |  | "You<br>have<br>passed<br>Questio<br>ne One" |
| 18 |  |  |    | 1 |  |      |  |                                              |
| 19 |  |  |    |   |  |      |  |                                              |

| Data Set 2  |                        |       |            |        |              |                                        |               |               |                        |
|-------------|------------------------|-------|------------|--------|--------------|----------------------------------------|---------------|---------------|------------------------|
| Line<br>No. | correct<br>_answ<br>er | lives | answe<br>r | streak | lives ><br>0 | answe<br>r =<br>correct<br>_answ<br>er | lives<br><= 0 | lives<br>== 1 | Expect<br>ed<br>Output |
| 1           | 30                     | 5     |            | 0      |              |                                        |               |               |                        |
| 2           |                        |       |            |        | TRUE         |                                        |               |               |                        |
| 3           |                        |       |            |        |              |                                        |               |               | "Quest<br>ion<br>One"  |
| 4           |                        |       |            |        |              |                                        |               |               | "5+25"                 |
| 5           |                        |       | 29         |        |              |                                        |               |               |                        |
| 6           |                        |       |            |        |              | FALSE                                  |               |               |                        |
| 7           |                        |       |            | 0      |              |                                        |               |               |                        |
| 8           |                        | 4     |            |        |              |                                        |               |               |                        |



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|    |  |   |    |   |      |       |  |       |                                                       |
|----|--|---|----|---|------|-------|--|-------|-------------------------------------------------------|
| 12 |  |   |    |   |      |       |  |       |                                                       |
| 13 |  |   |    |   |      |       |  |       | “You have lost one life. You have 3 lives remaining.” |
| 14 |  |   |    |   |      |       |  |       |                                                       |
| 20 |  |   |    |   |      |       |  |       |                                                       |
| 2  |  |   |    |   | True |       |  |       |                                                       |
| 3  |  |   |    |   |      |       |  |       | “Question One”                                        |
| 4  |  |   |    |   |      |       |  |       | “5+25”                                                |
| 5  |  |   | 29 |   |      |       |  |       |                                                       |
| 6  |  |   |    |   |      | FALSE |  |       |                                                       |
| 7  |  |   |    | 0 |      |       |  |       |                                                       |
| 8  |  | 4 |    |   |      |       |  |       |                                                       |
| 9  |  |   |    |   |      |       |  |       | “Incorrect Answer! Streak Lost!”                      |
| 10 |  |   |    |   |      |       |  | FALSE |                                                       |
| 12 |  |   |    |   |      |       |  |       |                                                       |
| 13 |  |   |    |   |      |       |  |       | “You have lost one life. You have 2                   |

|    |  |   |    |   |      |       |  |      |                                   |
|----|--|---|----|---|------|-------|--|------|-----------------------------------|
|    |  |   |    |   |      |       |  |      | lives remain ing.                 |
| 14 |  |   |    |   |      |       |  |      |                                   |
| 20 |  |   |    |   |      |       |  |      |                                   |
| 2  |  |   |    |   | True |       |  |      |                                   |
| 3  |  |   |    |   |      |       |  |      | “Quest ion One”                   |
| 4  |  |   |    |   |      |       |  |      | “5+25”                            |
| 5  |  |   | 29 |   |      |       |  |      |                                   |
| 6  |  |   |    |   |      | FALSE |  |      |                                   |
| 7  |  |   |    | 0 |      |       |  |      |                                   |
| 8  |  | 4 |    |   |      |       |  |      |                                   |
| 9  |  |   |    |   |      |       |  |      | “Incorr ect Answer! Streak Lost!” |
| 10 |  |   |    |   |      |       |  | TRUE |                                   |
| 11 |  |   |    |   |      |       |  |      | “You have 1 life remain ing”      |
| 2  |  |   |    |   | True |       |  |      |                                   |
| 3  |  |   |    |   |      |       |  |      | “Quest ion One”                   |
| 4  |  |   |    |   |      |       |  |      | “5+25”                            |
| 5  |  |   | 29 |   |      |       |  |      |                                   |
| 6  |  |   |    |   |      | FALSE |  |      |                                   |
| 7  |  |   |    | 0 |      |       |  |      |                                   |

|    |  |   |  |  |       |  |      |       |                                                       |
|----|--|---|--|--|-------|--|------|-------|-------------------------------------------------------|
| 8  |  | 4 |  |  |       |  |      |       |                                                       |
| 9  |  |   |  |  |       |  |      |       | "Incorrect Answer! Streak Lost!"                      |
| 10 |  |   |  |  |       |  |      | FALSE |                                                       |
| 12 |  |   |  |  |       |  |      |       |                                                       |
| 13 |  |   |  |  |       |  |      |       | "You have lost one life. You have 0 lives remaining." |
| 14 |  |   |  |  |       |  |      |       |                                                       |
| 20 |  |   |  |  | FALSE |  |      |       |                                                       |
| 21 |  |   |  |  |       |  |      |       |                                                       |
| 22 |  |   |  |  |       |  | TRUE |       |                                                       |
| 23 |  |   |  |  |       |  |      |       | "Game over"                                           |
| 24 |  |   |  |  |       |  |      |       |                                                       |

## BossQ1()

### Test Data

| Data Set | lives | streak | correct_answer | calculator | choice | answer | Expected Output   |
|----------|-------|--------|----------------|------------|--------|--------|-------------------|
| 1        | 5     | 5      | 7              | TRUE       | C      | 7      | "You have passed" |



|    |  |   |  |           |    |  |  |          |          |          |  |  |                                                           |
|----|--|---|--|-----------|----|--|--|----------|----------|----------|--|--|-----------------------------------------------------------|
|    |  |   |  |           |    |  |  |          |          |          |  |  | esti<br>on<br>1"                                          |
| 4  |  |   |  |           |    |  |  |          |          |          |  |  | "2(1<br>+3)<br>-<br>7/(4<br>+3)<br>+<br>(1+3<br>) x<br>0" |
| 5  |  |   |  |           | "" |  |  |          |          |          |  |  |                                                           |
| 6  |  |   |  |           |    |  |  | TR<br>UE |          |          |  |  |                                                           |
| 7  |  |   |  |           |    |  |  |          |          |          |  |  | 'Inp<br>ut A<br>to<br>ans<br>wer'                         |
| 8  |  |   |  |           |    |  |  |          |          |          |  |  | 'Inp<br>ut C<br>to<br>use<br>calc<br>ulat<br>or'          |
| 9  |  |   |  |           | c  |  |  |          |          |          |  |  |                                                           |
| 10 |  |   |  |           | C  |  |  |          |          |          |  |  |                                                           |
| 11 |  |   |  |           |    |  |  |          | TR<br>UE | TR<br>UE |  |  |                                                           |
| 12 |  |   |  | FAL<br>SE |    |  |  |          |          |          |  |  |                                                           |
| 13 |  | 6 |  |           |    |  |  |          |          |          |  |  |                                                           |
| 14 |  |   |  |           |    |  |  |          |          |          |  |  | You<br>hav<br>e<br>pas<br>sed<br>usin                     |

[illegible][illegible]







|    |  |   |  |  |  |   |  |  |       |  |      |  |       |                                |
|----|--|---|--|--|--|---|--|--|-------|--|------|--|-------|--------------------------------|
| 11 |  |   |  |  |  |   |  |  | FALSE |  |      |  |       |                                |
| 20 |  |   |  |  |  |   |  |  |       |  | TRUE |  |       |                                |
| 21 |  |   |  |  |  |   |  |  |       |  |      |  |       | “Enter your answer”            |
| 22 |  |   |  |  |  | 7 |  |  |       |  |      |  |       |                                |
| 23 |  |   |  |  |  |   |  |  |       |  |      |  | FALSE |                                |
| 32 |  |   |  |  |  |   |  |  |       |  |      |  |       |                                |
| 33 |  |   |  |  |  |   |  |  |       |  |      |  |       | “Correct!”                     |
| 34 |  |   |  |  |  |   |  |  |       |  |      |  |       | “You have passed Question One” |
| 35 |  | 2 |  |  |  |   |  |  |       |  |      |  |       |                                |
| 36 |  |   |  |  |  |   |  |  |       |  |      |  |       |                                |

Data Set 4



|    |   |   |  |  |   |   |  |  |               |  |          |  |          |  |                                                     |
|----|---|---|--|--|---|---|--|--|---------------|--|----------|--|----------|--|-----------------------------------------------------|
| 8  |   |   |  |  |   |   |  |  |               |  |          |  |          |  | “Input<br>C<br>to<br>use<br>cal<br>cul<br>ator<br>” |
| 9  |   |   |  |  | A |   |  |  |               |  |          |  |          |  |                                                     |
| 10 |   |   |  |  | A |   |  |  |               |  |          |  |          |  |                                                     |
| 11 |   |   |  |  |   |   |  |  | FA<br>LS<br>E |  |          |  |          |  |                                                     |
| 20 |   |   |  |  |   |   |  |  |               |  | TR<br>UE |  |          |  |                                                     |
| 21 |   |   |  |  |   |   |  |  |               |  |          |  |          |  | “En<br>ter<br>you<br>r<br>ans<br>wer<br>”           |
| 22 |   |   |  |  |   | 6 |  |  |               |  |          |  |          |  |                                                     |
| 23 |   |   |  |  |   |   |  |  |               |  |          |  | TR<br>UE |  |                                                     |
| 24 |   | 0 |  |  |   |   |  |  |               |  |          |  |          |  |                                                     |
| 25 | 4 |   |  |  |   |   |  |  |               |  |          |  |          |  |                                                     |
| 26 |   |   |  |  |   |   |  |  |               |  |          |  |          |  | “Inc<br>orr<br>ect<br>Ans<br>wer<br>!               |

|    |  |  |  |  |  |  |  |  |  |  |  |  |  |               |                                                                                                       |
|----|--|--|--|--|--|--|--|--|--|--|--|--|--|---------------|-------------------------------------------------------------------------------------------------------|
|    |  |  |  |  |  |  |  |  |  |  |  |  |  |               | Str<br>eak<br>Los<br>t!                                                                               |
| 27 |  |  |  |  |  |  |  |  |  |  |  |  |  | FA<br>LS<br>E |                                                                                                       |
| 29 |  |  |  |  |  |  |  |  |  |  |  |  |  |               |                                                                                                       |
| 30 |  |  |  |  |  |  |  |  |  |  |  |  |  |               | “Yo<br>u<br>hav<br>e<br>lost<br>one<br>life.<br>You<br>hav<br>e 4<br>live<br>s<br>rem<br>aini<br>ng.” |

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|    |   |   |  |  |   |   |  |  |               |  |          |  |          |          |                                                                   |
|----|---|---|--|--|---|---|--|--|---------------|--|----------|--|----------|----------|-------------------------------------------------------------------|
| 10 |   |   |  |  | A |   |  |  |               |  |          |  |          |          |                                                                   |
| 11 |   |   |  |  |   |   |  |  | FA<br>LS<br>E |  |          |  |          |          |                                                                   |
| 20 |   |   |  |  |   |   |  |  |               |  | TR<br>UE |  |          |          |                                                                   |
| 21 |   |   |  |  |   |   |  |  |               |  |          |  |          |          | “En<br>ter<br>you<br>r<br>ans<br>wer<br>”                         |
| 22 |   |   |  |  |   | 6 |  |  |               |  |          |  |          |          |                                                                   |
| 23 |   |   |  |  |   |   |  |  |               |  |          |  | TR<br>UE |          |                                                                   |
| 24 |   | 0 |  |  |   |   |  |  |               |  |          |  |          |          |                                                                   |
| 25 | 1 |   |  |  |   |   |  |  |               |  |          |  |          |          |                                                                   |
| 26 |   |   |  |  |   |   |  |  |               |  |          |  |          |          | “Inc<br>orr<br>ect<br>Ans<br>wer<br>!<br>Str<br>eak<br>Los<br>t!” |
| 27 |   |   |  |  |   |   |  |  |               |  |          |  |          | TR<br>UE |                                                                   |
| 28 |   |   |  |  |   |   |  |  |               |  |          |  |          |          | “Yo<br>u<br>hav<br>e 1                                            |

|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                 |
|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-----------------|
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | life remaining” |
|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-----------------|

| Data Set 6 |       |        |                |            |        |        |           |                      |              |                   |              |            |                          |           |                                |
|------------|-------|--------|----------------|------------|--------|--------|-----------|----------------------|--------------|-------------------|--------------|------------|--------------------------|-----------|--------------------------------|
| Line No.   | lives | streak | correct_answer | calculator | choice | answer | lives > 0 | choice != 'A' or 'C' | choice = 'C' | calculator = True | choice = 'A' | lives <= 0 | answer != correct_answer | lives = 1 | Output                         |
| 1          | 1     | 1      | 7              | FALSE      |        |        |           |                      |              |                   |              |            |                          |           |                                |
| 2          |       |        |                |            |        |        | TRUE      |                      |              |                   |              |            |                          |           |                                |
| 3          |       |        |                |            |        |        |           |                      |              |                   |              |            |                          |           | “Question 1”                   |
| 4          |       |        |                |            |        |        |           |                      |              |                   |              |            |                          |           | “2(1+3) - 7/(4+3) + (1+3) x 0” |





[illegible]

|    |  |  |  |  |  |  |  |  |  |  |  |      |  |  |              |
|----|--|--|--|--|--|--|--|--|--|--|--|------|--|--|--------------|
| 39 |  |  |  |  |  |  |  |  |  |  |  | TRUE |  |  |              |
| 40 |  |  |  |  |  |  |  |  |  |  |  |      |  |  | "Game Over." |
| 41 |  |  |  |  |  |  |  |  |  |  |  |      |  |  |              |

Tmain()

Test Data

| Data Set | begin | username | username_confirmed | lives | streak | start | questions_finished | calculator | end | continue | Expected Output        |
|----------|-------|----------|--------------------|-------|--------|-------|--------------------|------------|-----|----------|------------------------|
| 1        | "     | Will     | Y                  | 5     | 7      | 0     | FALSE              | TRUE       | 120 | TRUE     | "GAME COMPLETED"       |
| 2        | "     | Will     | Y                  | 3     | 2      | 0     |                    |            |     |          | "You now have 4 lives" |
| 3        | "     | Will     | Y                  | 3     | 1      | 0     | FALSE              | FALSE      | 120 | TRUE     | "GAME COMPLETED"       |
| 4        | "     | Will     | Y                  | 0     | 0      | 0     | FALSE              |            |     |          | Program is exited      |
| 5        | "     | Will     | Y                  | 3     | 2      | 0     | FALSE              | FALSE      |     |          | Program is exited      |



|    |  |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                                                                                                        |
|----|--|--|--|--|--|--|--|--|--|------------------|--|--|--|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |  |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  | rie<br>s<br>of<br>qu<br>es<br>tio<br>ns<br>to<br>sa<br>ve<br>th<br>e<br>la<br>nd<br>fro<br>m<br>th<br>e<br>tre<br>ac<br>he<br>ro<br>us<br>M<br>at<br>h<br>In<br>qu<br>isit<br>or!<br>" |
| 8  |  |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  | "G<br>oo<br>d<br>lu<br>ck<br>br<br>av<br>e<br>so<br>ul.<br>.."                                                                                                                         |
| 9  |  |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                                                                                                        |
| 10 |  |  |  |  |  |  |  |  |  | T<br>R<br>U<br>E |  |  |  |  |  |  |                                                                                                                                                                                        |

|    |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                        |
|----|---|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--------------------------------------------------------------------------------------------------------|
| 11 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "P<br>re<br>ss<br>E<br>N<br>T<br>E<br>R<br>to<br>st<br>art<br>"                                        |
| 12 | ' |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                        |
| 17 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                        |
| 18 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                        |
| 19 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | '                                                                                                      |
| 20 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "L<br>oa<br>di<br>ng<br>"                                                                              |
| 21 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | '                                                                                                      |
| 22 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                        |
| 23 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                        |
| 24 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "S<br>ho<br>wi<br>ng<br>pr<br>ev<br>io<br>us<br>sc<br>or<br>es<br>. If<br>nil<br>, th<br>er<br>e<br>ar |

|    |  |  |  |  |  |  |  |  |  |  |  |        |  |  |  |  |  |                                                                                                                                                                                                                      |
|----|--|--|--|--|--|--|--|--|--|--|--|--------|--|--|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |  |  |  |  |  |  |  |  |  |  |  |        |  |  |  |  |  | e<br>no<br>pr<br>ev<br>io<br>us<br>sc<br>or<br>es<br>.<br>St<br>ati<br>sti<br>cs<br>wil<br>l<br>on<br>ly<br>be<br>re<br>co<br>rd<br>ed<br>if<br>th<br>e<br>en<br>tir<br>e<br>ru<br>n<br>is<br>fin<br>is<br>he<br>d." |
| 25 |  |  |  |  |  |  |  |  |  |  |  |        |  |  |  |  |  |                                                                                                                                                                                                                      |
| 26 |  |  |  |  |  |  |  |  |  |  |  |        |  |  |  |  |  |                                                                                                                                                                                                                      |
| 27 |  |  |  |  |  |  |  |  |  |  |  |        |  |  |  |  |  |                                                                                                                                                                                                                      |
| 28 |  |  |  |  |  |  |  |  |  |  |  |        |  |  |  |  |  |                                                                                                                                                                                                                      |
| 29 |  |  |  |  |  |  |  |  |  |  |  |        |  |  |  |  |  |                                                                                                                                                                                                                      |
| 30 |  |  |  |  |  |  |  |  |  |  |  |        |  |  |  |  |  | "                                                                                                                                                                                                                    |
| 31 |  |  |  |  |  |  |  |  |  |  |  | T<br>R |  |  |  |  |  |                                                                                                                                                                                                                      |

|    |  |          |   |  |  |  |  |  |  |  |  |        |  |  |  |  |  |                                                                                        |
|----|--|----------|---|--|--|--|--|--|--|--|--|--------|--|--|--|--|--|----------------------------------------------------------------------------------------|
|    |  |          |   |  |  |  |  |  |  |  |  | U<br>E |  |  |  |  |  |                                                                                        |
| 32 |  | Wi<br>ll |   |  |  |  |  |  |  |  |  |        |  |  |  |  |  |                                                                                        |
| 33 |  |          |   |  |  |  |  |  |  |  |  |        |  |  |  |  |  | "C<br>on<br>fir<br>m<br>us<br>er<br>na<br>me<br>as<br>:<br>Wi<br>ll?<br>(Y<br>/N<br>)" |
| 34 |  |          | Y |  |  |  |  |  |  |  |  |        |  |  |  |  |  |                                                                                        |
| 35 |  |          |   |  |  |  |  |  |  |  |  |        |  |  |  |  |  |                                                                                        |
| 36 |  |          |   |  |  |  |  |  |  |  |  |        |  |  |  |  |  | "U<br>se<br>rn<br>a<br>me<br>is<br>co<br>nfi<br>rm<br>ed<br>as<br>Wi<br>ll"            |
| 37 |  |          |   |  |  |  |  |  |  |  |  |        |  |  |  |  |  | "Ti<br>me<br>st<br>art<br>s<br>no<br>w"                                                |



|    |  |  |  |   |   |   |                  |  |  |  |  |  |                  |                  |               |  |  |                                                                |
|----|--|--|--|---|---|---|------------------|--|--|--|--|--|------------------|------------------|---------------|--|--|----------------------------------------------------------------|
| 38 |  |  |  |   |   | 0 |                  |  |  |  |  |  |                  |                  |               |  |  |                                                                |
| 39 |  |  |  |   |   |   |                  |  |  |  |  |  | T<br>R<br>U<br>E | T<br>R<br>U<br>E |               |  |  |                                                                |
| 40 |  |  |  | 5 | 7 |   |                  |  |  |  |  |  |                  |                  |               |  |  |                                                                |
| 41 |  |  |  |   |   |   |                  |  |  |  |  |  |                  |                  |               |  |  | "S<br>tat<br>s!"                                               |
| 42 |  |  |  |   |   |   |                  |  |  |  |  |  |                  |                  |               |  |  | "Y<br>ou<br>r<br>cu<br>rre<br>nt<br>str<br>ea<br>k<br>is<br>5" |
| 43 |  |  |  |   |   |   |                  |  |  |  |  |  |                  |                  |               |  |  | "Li<br>ve<br>s<br>re<br>m<br>ai<br>ni<br>ng<br>:<br>5"         |
| 44 |  |  |  |   |   |   | T<br>R<br>U<br>E |  |  |  |  |  |                  |                  |               |  |  |                                                                |
| 39 |  |  |  |   |   |   |                  |  |  |  |  |  | T<br>R<br>U<br>E | FA<br>LS<br>E    |               |  |  |                                                                |
| 45 |  |  |  |   |   |   |                  |  |  |  |  |  |                  |                  |               |  |  |                                                                |
| 46 |  |  |  |   |   |   |                  |  |  |  |  |  |                  |                  | FA<br>LS<br>E |  |  |                                                                |

[illegible]

|    |  |  |  |  |  |  |  |             |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                        |
|----|--|--|--|--|--|--|--|-------------|--|--|--|--|--|--|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |  |  |  |  |  |  |  | R<br>U<br>E |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                        |
| 56 |  |  |  |  |  |  |  |             |  |  |  |  |  |  |  |  |  | "S<br>in<br>ce<br>yo<br>u<br>ha<br>ve<br>a<br>str<br>ea<br>k<br>of<br>5,<br>yo<br>u<br>ha<br>ve<br>re<br>ce<br>iv<br>ed<br>a<br>C<br>AL<br>C<br>U<br>LA<br>T<br>O<br>R.<br>Us<br>e<br>it<br>in<br>th<br>e<br>bo<br>ss<br>fig<br>ht<br>to<br>im<br>m<br>ed<br>iat<br>el |

|    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                                        |
|----|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------|
|    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | y<br>an<br>sw<br>er<br>an<br>y<br>qu<br>es<br>tio<br>n!"                                                                               |
| 59 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                                        |
| 60 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "L<br>oa<br>di<br>ng<br>B<br>O<br>S<br>S<br>FI<br>G<br>H<br>T"                                                                         |
| 61 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 'H<br>er<br>e<br>lie<br>s<br>th<br>e<br>M<br>at<br>h<br>In<br>qu<br>isit<br>or,<br>th<br>e<br>fin<br>al<br>bo<br>ss<br>. Be<br>w<br>ar |

|    |  |  |  |   |   |  |               |               |  |  |  |  |                  |                  |  |  |  |                                                                                 |
|----|--|--|--|---|---|--|---------------|---------------|--|--|--|--|------------------|------------------|--|--|--|---------------------------------------------------------------------------------|
|    |  |  |  |   |   |  |               |               |  |  |  |  |                  |                  |  |  |  | e<br>of<br>hi<br>s<br>de<br>ad<br>ly<br>qu<br>es<br>tio<br>ns<br>..'            |
| 62 |  |  |  |   |   |  | FA<br>LS<br>E |               |  |  |  |  |                  |                  |  |  |  |                                                                                 |
| 63 |  |  |  |   |   |  |               |               |  |  |  |  | T<br>R<br>U<br>E | T<br>R<br>U<br>E |  |  |  |                                                                                 |
| 64 |  |  |  | 6 | 8 |  |               | FA<br>LS<br>E |  |  |  |  |                  |                  |  |  |  |                                                                                 |
| 65 |  |  |  |   |   |  |               |               |  |  |  |  |                  |                  |  |  |  | "S<br>tat<br>s!"                                                                |
| 66 |  |  |  |   |   |  |               |               |  |  |  |  |                  |                  |  |  |  | "Y<br>ou<br>r<br>cu<br>rre<br>nt<br>an<br>sw<br>er<br>st<br>ea<br>k<br>is<br>8" |
| 67 |  |  |  |   |   |  |               |               |  |  |  |  |                  |                  |  |  |  | "Li<br>ve<br>s<br>re<br>m                                                       |

|    |  |  |  |  |  |  |                  |  |         |                  |  |                  |                       |                       |  |  |  |                                                                                                              |
|----|--|--|--|--|--|--|------------------|--|---------|------------------|--|------------------|-----------------------|-----------------------|--|--|--|--------------------------------------------------------------------------------------------------------------|
|    |  |  |  |  |  |  |                  |  |         |                  |  |                  |                       |                       |  |  |  | ai<br>ni<br>ng<br>:<br>6"                                                                                    |
| 68 |  |  |  |  |  |  |                  |  |         | T<br>R<br>U<br>E |  |                  |                       |                       |  |  |  |                                                                                                              |
| 69 |  |  |  |  |  |  | T<br>R<br>U<br>E |  |         |                  |  |                  |                       |                       |  |  |  |                                                                                                              |
| 63 |  |  |  |  |  |  |                  |  |         |                  |  | T<br>R<br>U<br>E | F<br>A<br>L<br>S<br>E |                       |  |  |  |                                                                                                              |
| 70 |  |  |  |  |  |  |                  |  |         |                  |  |                  |                       |                       |  |  |  |                                                                                                              |
| 71 |  |  |  |  |  |  |                  |  |         |                  |  |                  |                       | F<br>A<br>L<br>S<br>E |  |  |  |                                                                                                              |
| 73 |  |  |  |  |  |  |                  |  |         |                  |  |                  |                       |                       |  |  |  |                                                                                                              |
| 74 |  |  |  |  |  |  |                  |  | 12<br>0 |                  |  |                  |                       |                       |  |  |  |                                                                                                              |
| 75 |  |  |  |  |  |  |                  |  |         |                  |  |                  |                       |                       |  |  |  | Yo<br>u<br>ha<br>ve<br>de<br>fe<br>at<br>ed<br>th<br>e<br>M<br>at<br>h<br>In<br>qu<br>isit<br>or!<br>Yo<br>u |

|    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                               |
|----|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-----------------------------------------------------------------------------------------------|
|    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | lie<br>th<br>e<br>Ki<br>ng<br>of<br>th<br>e<br>la<br>nd<br>.'                                 |
| 76 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "Y<br>ou<br>r<br>ti<br>me<br>was<br>2<br>mi<br>nu<br>te<br>s."                                |
| 77 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "Y<br>ou<br>r<br>st<br>ati<br>sti<br>cs<br>ha<br>ve<br>be<br>en<br>re<br>co<br>rd<br>ed<br>." |
| 78 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "G<br>A<br>M<br>E<br>C<br>O<br>M                                                              |





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|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                            |
|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|----------------------------------------------------------------------------------------------------------------------------|
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | wing previous scores. If nil, there are no previous scores. Statistics will only be recorded if the entire run is finished |
|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|----------------------------------------------------------------------------------------------------------------------------|

|    |  |          |   |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                        |
|----|--|----------|---|--|--|--|--|--|--|--|------------------|--|--|--|--|--|--|----------------------------------------------------------------------------------------|
|    |  |          |   |  |  |  |  |  |  |  |                  |  |  |  |  |  |  | he<br>d."                                                                              |
| 25 |  |          |   |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                        |
| 26 |  |          |   |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                        |
| 27 |  |          |   |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                        |
| 28 |  |          |   |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                        |
| 29 |  |          |   |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                        |
| 30 |  |          |   |  |  |  |  |  |  |  |                  |  |  |  |  |  |  | ,                                                                                      |
| 31 |  |          |   |  |  |  |  |  |  |  | T<br>R<br>U<br>E |  |  |  |  |  |  |                                                                                        |
| 32 |  | Wi<br>ll |   |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                        |
| 33 |  |          |   |  |  |  |  |  |  |  |                  |  |  |  |  |  |  | "C<br>on<br>fir<br>m<br>us<br>er<br>na<br>me<br>as<br>:<br>Wi<br>ll?<br>(Y<br>/N<br>)" |
| 34 |  |          | Y |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                        |
| 35 |  |          |   |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                        |
| 36 |  |          |   |  |  |  |  |  |  |  |                  |  |  |  |  |  |  | "U<br>se<br>rn<br>a<br>me<br>is                                                        |

[illegible]

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|    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                          |
|----|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|----------------------------------------------------------|
|    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | w<br>h<br>a<br>v<br>e<br>4<br>l<br>i<br>v<br>e<br>s<br>" |
| 53 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                          |

| Data Set 3      |               |                      |                                            |           |                |           |                                                |                            |         |                            |                          |                                                             |                     |                                                                      |                      |                      |                      |                                            |
|-----------------|---------------|----------------------|--------------------------------------------|-----------|----------------|-----------|------------------------------------------------|----------------------------|---------|----------------------------|--------------------------|-------------------------------------------------------------|---------------------|----------------------------------------------------------------------|----------------------|----------------------|----------------------|--------------------------------------------|
| 1               | 2             | 3                    | 4                                          | 5         | 6              | 7         | 8                                              | 9                          | 10      | 11                         | 12                       | 13                                                          | 14                  | 15                                                                   | 16                   | 17                   | 18                   | 19                                         |
| Line<br>N<br>o. | be<br>gi<br>n | us<br>er<br>na<br>me | us<br>er<br>na<br>me_<br>co<br>nfi<br>rmed | liv<br>es | str<br>ea<br>k | st<br>art | qu<br>es<br>tio<br>ns_<br>fi<br>ni<br>sh<br>ed | ca<br>lc<br>ul<br>at<br>or | en<br>d | co<br>nti<br>nu<br>eo<br>n | be<br>gi<br>n<br>!=<br>" | us<br>er<br>na<br>me_<br>co<br>nfi<br>rmed<br>!=<br>"Y<br>" | liv<br>es<br>><br>0 | qu<br>es<br>tio<br>ns_<br>fi<br>ni<br>sh<br>ed<br>=<br>Fa<br>ls<br>e | liv<br>es<br><=<br>0 | liv<br>es<br>>=<br>2 | liv<br>es<br>>=<br>5 | Ex<br>pe<br>ct<br>ed<br>O<br>ut<br>pu<br>t |
| 1               | 0             |                      |                                            |           |                |           |                                                |                            |         |                            |                          |                                                             |                     |                                                                      |                      |                      |                      |                                            |
| 2               |               | "                    |                                            |           |                |           |                                                |                            |         |                            |                          |                                                             |                     |                                                                      |                      |                      |                      |                                            |
| 3               |               |                      | N                                          |           |                |           |                                                |                            |         |                            |                          |                                                             |                     |                                                                      |                      |                      |                      |                                            |
| 4               |               |                      |                                            | 5         |                |           |                                                |                            |         |                            |                          |                                                             |                     |                                                                      |                      |                      |                      |                                            |
| 5               |               |                      |                                            |           | 0              |           |                                                |                            |         |                            |                          |                                                             |                     |                                                                      |                      |                      |                      |                                            |
| 6               |               |                      |                                            |           |                |           |                                                |                            |         |                            |                          |                                                             |                     |                                                                      |                      |                      |                      | " Wel<br>co<br>me<br>to<br>Mat<br>h<br>Que |

|   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                          |
|---|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | st!<br>"                                                                                                                                                                                                                 |
| 7 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "A<br>ns<br>w<br>er<br>a<br>se<br>rie<br>s<br>of<br>qu<br>es<br>tio<br>ns<br>to<br>sa<br>ve<br>th<br>e<br>la<br>nd<br>fro<br>m<br>th<br>e<br>tre<br>ac<br>he<br>ro<br>us<br>M<br>at<br>h<br>In<br>qu<br>isit<br>or!<br>" |
| 8 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "G<br>oo<br>d<br>lu<br>ck<br>br<br>av<br>e<br>so                                                                                                                                                                         |



|    |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                                                                 |
|----|---|--|--|--|--|--|--|--|--|------------------|--|--|--|--|--|--|--|-----------------------------------------------------------------|
|    |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  | ul.<br>.."                                                      |
| 9  |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                                                                 |
| 10 |   |  |  |  |  |  |  |  |  | T<br>R<br>U<br>E |  |  |  |  |  |  |  |                                                                 |
| 11 |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  | "P<br>re<br>ss<br>E<br>N<br>T<br>E<br>R<br>to<br>st<br>art<br>" |
| 12 | " |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                                                                 |
| 17 |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                                                                 |
| 18 |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                                                                 |
| 19 |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  | "                                                               |
| 20 |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  | "L<br>oa<br>di<br>ng<br>"                                       |
| 21 |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  | "                                                               |
| 22 |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                                                                 |
| 23 |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                                                                 |
| 24 |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  | "S<br>ho<br>wi<br>ng<br>pr<br>ev<br>io<br>us                    |

[illegible]

|    |  |          |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |                                                                                        |
|----|--|----------|---|--|--|--|--|--|--|--|--|------------------|--|--|--|--|--|----------------------------------------------------------------------------------------|
| 27 |  |          |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |                                                                                        |
| 28 |  |          |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |                                                                                        |
| 29 |  |          |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |                                                                                        |
| 30 |  |          |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  | '                                                                                      |
| 31 |  |          |   |  |  |  |  |  |  |  |  | T<br>R<br>U<br>E |  |  |  |  |  |                                                                                        |
| 32 |  | Wi<br>ll |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |                                                                                        |
| 33 |  |          |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  | "C<br>on<br>fir<br>m<br>us<br>er<br>na<br>me<br>as<br>:<br>Wi<br>ll?<br>(Y<br>/N<br>)" |
| 34 |  |          | Y |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |                                                                                        |
| 35 |  |          |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |                                                                                        |
| 36 |  |          |   |  |  |  |  |  |  |  |  |                  |  |  |  |  |  | "U<br>se<br>rn<br>a<br>me<br>is<br>co<br>nfi<br>rm<br>ed<br>as<br>Wi<br>ll"            |

[illegible]

|    |  |  |  |  |  |  |   |                       |  |  |  |  |                  |                       |                       |                       |  |                                                                                     |
|----|--|--|--|--|--|--|---|-----------------------|--|--|--|--|------------------|-----------------------|-----------------------|-----------------------|--|-------------------------------------------------------------------------------------|
|    |  |  |  |  |  |  | E |                       |  |  |  |  |                  |                       |                       |                       |  |                                                                                     |
| 39 |  |  |  |  |  |  |   |                       |  |  |  |  | T<br>R<br>U<br>E | F<br>A<br>L<br>S<br>E |                       |                       |  |                                                                                     |
| 45 |  |  |  |  |  |  |   |                       |  |  |  |  |                  |                       |                       |                       |  |                                                                                     |
| 46 |  |  |  |  |  |  |   |                       |  |  |  |  |                  | F<br>A<br>L<br>S<br>E |                       |                       |  |                                                                                     |
| 48 |  |  |  |  |  |  |   |                       |  |  |  |  |                  |                       |                       |                       |  |                                                                                     |
| 49 |  |  |  |  |  |  |   |                       |  |  |  |  |                  |                       | F<br>A<br>L<br>S<br>E |                       |  |                                                                                     |
| 53 |  |  |  |  |  |  |   |                       |  |  |  |  |                  |                       |                       |                       |  |                                                                                     |
| 54 |  |  |  |  |  |  |   |                       |  |  |  |  |                  |                       |                       | F<br>A<br>L<br>S<br>E |  |                                                                                     |
| 57 |  |  |  |  |  |  |   |                       |  |  |  |  |                  |                       |                       |                       |  |                                                                                     |
| 58 |  |  |  |  |  |  |   | F<br>A<br>L<br>S<br>E |  |  |  |  |                  |                       |                       |                       |  |                                                                                     |
| 59 |  |  |  |  |  |  |   |                       |  |  |  |  |                  |                       |                       |                       |  |                                                                                     |
| 60 |  |  |  |  |  |  |   |                       |  |  |  |  |                  |                       |                       |                       |  | “L<br>o<br>a<br>d<br>i<br>n<br>g<br>B<br>O<br>O<br>S<br>S<br>F<br>I<br>G<br>H<br>T” |
| 61 |  |  |  |  |  |  |   |                       |  |  |  |  |                  |                       |                       |                       |  | “H<br>e<br>r<br>e<br>l<br>i<br>e                                                    |



[illegible]

[illegible]



|    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                     |
|----|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|---------------------|
|    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | have been recorded" |
| 78 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "GAME COMPLETED"    |
| 79 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                     |
| 80 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                     |
| 81 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                     |
| 82 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                     |
| 83 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                     |

| Data Set 4 |       |          |                    |       |        |       |                    |            |     |              |          |                        |         |                          |          |           |           |                 |
|------------|-------|----------|--------------------|-------|--------|-------|--------------------|------------|-----|--------------|----------|------------------------|---------|--------------------------|----------|-----------|-----------|-----------------|
| 1          | 2     | 3        | 4                  | 5     | 6      | 7     | 8                  | 9          | 10  | 11           | 12       | 13                     | 14      | 15                       | 16       | 17        | 18        | 19              |
| Line No.   | begin | username | username_confirmed | lives | streak | start | questions_finished | calculator | end | continuation | begin!=" | username_confirmed!="Y | lives>0 | questions_finished=False | lives<=0 | streak>=2 | streak>=5 | Expected Output |

|   |   |   |   |   |   |  |  |  |  |  |  |   |  |   |  |  |  |                                                          |
|---|---|---|---|---|---|--|--|--|--|--|--|---|--|---|--|--|--|----------------------------------------------------------|
|   |   |   |   |   |   |  |  |  |  |  |  | " |  | e |  |  |  |                                                          |
| 1 | 0 |   |   |   |   |  |  |  |  |  |  |   |  |   |  |  |  |                                                          |
| 2 |   | ' |   |   |   |  |  |  |  |  |  |   |  |   |  |  |  |                                                          |
| 3 |   |   | N |   |   |  |  |  |  |  |  |   |  |   |  |  |  |                                                          |
| 4 |   |   |   | 5 |   |  |  |  |  |  |  |   |  |   |  |  |  |                                                          |
| 5 |   |   |   |   | 0 |  |  |  |  |  |  |   |  |   |  |  |  |                                                          |
| 6 |   |   |   |   |   |  |  |  |  |  |  |   |  |   |  |  |  | " Wel come to Math Queue st!"                            |
| 7 |   |   |   |   |   |  |  |  |  |  |  |   |  |   |  |  |  | "Answer a series of questions to save the land from the" |

[illegible]

|    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                 |
|----|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                 |
| 20 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "L<br>oa<br>di<br>ng<br>"                                                                                                                                                                                       |
| 21 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "                                                                                                                                                                                                               |
| 22 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                 |
| 23 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                 |
| 24 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "S<br>ho<br>wi<br>ng<br>pr<br>ev<br>io<br>us<br>sc<br>or<br>es<br>. If<br>nil<br>, th<br>er<br>e<br>ar<br>e<br>no<br>pr<br>ev<br>io<br>us<br>sc<br>or<br>es<br>. St<br>ati<br>sti<br>cs<br>wil<br>l<br>on<br>ly |

|    |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                          |
|----|--|----------|--|--|--|--|--|--|--|--|------------------|--|--|--|--|--|--|----------------------------------------------------------------------------------------------------------|
|    |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  | be<br>re<br>co<br>rd<br>ed<br>if<br>th<br>e<br>en<br>tir<br>e<br>ru<br>n<br>is<br>fin<br>is<br>he<br>d." |
| 25 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                          |
| 26 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                          |
| 27 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                          |
| 28 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                          |
| 29 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                          |
| 30 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  | "                                                                                                        |
| 31 |  |          |  |  |  |  |  |  |  |  | T<br>R<br>U<br>E |  |  |  |  |  |  |                                                                                                          |
| 32 |  | Wi<br>ll |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                          |
| 33 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  | "C<br>on<br>fir<br>m<br>us<br>er<br>na<br>me<br>as<br>:<br>Wi                                            |

|    |  |  |   |   |   |   |  |  |  |  |  |  |                  |                  |  |  |  |                                                                                 |
|----|--|--|---|---|---|---|--|--|--|--|--|--|------------------|------------------|--|--|--|---------------------------------------------------------------------------------|
|    |  |  |   |   |   |   |  |  |  |  |  |  |                  |                  |  |  |  | II?<br>(Y<br>/N<br>)"                                                           |
| 34 |  |  | Y |   |   |   |  |  |  |  |  |  |                  |                  |  |  |  |                                                                                 |
| 35 |  |  |   |   |   |   |  |  |  |  |  |  |                  |                  |  |  |  |                                                                                 |
| 36 |  |  |   |   |   |   |  |  |  |  |  |  |                  |                  |  |  |  | "U<br>se<br>rn<br>a<br>m<br>e<br>is<br>co<br>nfi<br>rm<br>ed<br>as<br>Wi<br>ll" |
| 37 |  |  |   |   |   |   |  |  |  |  |  |  |                  |                  |  |  |  | "Ti<br>m<br>e<br>st<br>art<br>s<br>no<br>w"                                     |
| 38 |  |  |   |   |   | 0 |  |  |  |  |  |  |                  |                  |  |  |  |                                                                                 |
| 39 |  |  |   |   |   |   |  |  |  |  |  |  | T<br>R<br>U<br>E | T<br>R<br>U<br>E |  |  |  |                                                                                 |
| 40 |  |  |   | 0 | 0 |   |  |  |  |  |  |  |                  |                  |  |  |  |                                                                                 |
| 41 |  |  |   |   |   |   |  |  |  |  |  |  |                  |                  |  |  |  | "S<br>tat<br>s!"                                                                |
| 42 |  |  |   |   |   |   |  |  |  |  |  |  |                  |                  |  |  |  | "Y<br>ou<br>r<br>cu<br>rre                                                      |

|    |  |  |  |  |  |  |                  |  |  |  |  |  |                       |                       |                  |  |  |                                                        |
|----|--|--|--|--|--|--|------------------|--|--|--|--|--|-----------------------|-----------------------|------------------|--|--|--------------------------------------------------------|
|    |  |  |  |  |  |  |                  |  |  |  |  |  |                       |                       |                  |  |  | nt<br>an<br>sw<br>er<br>str<br>ea<br>k<br>is<br>0"     |
| 43 |  |  |  |  |  |  |                  |  |  |  |  |  |                       |                       |                  |  |  | "Li<br>ve<br>s<br>re<br>m<br>ai<br>ni<br>ng<br>:<br>0" |
| 44 |  |  |  |  |  |  | T<br>R<br>U<br>E |  |  |  |  |  |                       |                       |                  |  |  |                                                        |
| 39 |  |  |  |  |  |  |                  |  |  |  |  |  | F<br>A<br>L<br>S<br>E | F<br>A<br>L<br>S<br>E |                  |  |  |                                                        |
| 45 |  |  |  |  |  |  |                  |  |  |  |  |  |                       |                       |                  |  |  |                                                        |
| 46 |  |  |  |  |  |  |                  |  |  |  |  |  |                       |                       | T<br>R<br>U<br>E |  |  |                                                        |
| 47 |  |  |  |  |  |  |                  |  |  |  |  |  |                       |                       |                  |  |  |                                                        |

| Data Set 5          |               |                      |                             |           |                |           |                              |                            |         |                            |                          |                             |                     |                              |                      |                           |                           |                           |
|---------------------|---------------|----------------------|-----------------------------|-----------|----------------|-----------|------------------------------|----------------------------|---------|----------------------------|--------------------------|-----------------------------|---------------------|------------------------------|----------------------|---------------------------|---------------------------|---------------------------|
| 1                   | 2             | 3                    | 4                           | 5         | 6              | 7         | 8                            | 9                          | 10      | 11                         | 12                       | 13                          | 14                  | 15                           | 16                   | 17                        | 18                        | 19                        |
| Li<br>ne<br>N<br>o. | be<br>gi<br>n | us<br>er<br>na<br>me | us<br>er<br>na<br>me_<br>e_ | liv<br>es | str<br>ea<br>k | st<br>art | qu<br>es<br>tio<br>ns_<br>fi | ca<br>lc<br>ul<br>at<br>or | en<br>d | co<br>nti<br>nu<br>eo<br>n | be<br>gi<br>n<br>!=<br>" | us<br>er<br>na<br>me_<br>e_ | liv<br>es<br>><br>0 | qu<br>es<br>tio<br>ns_<br>fi | liv<br>es<br><=<br>0 | str<br>ea<br>k<br>>=<br>2 | str<br>ea<br>k<br>>=<br>5 | Ex<br>pe<br>ct<br>ed<br>O |

|   |   |   | co<br>nfi<br>rm<br>ed |   |   |  | ni<br>sh<br>ed |  |  |  |  | co<br>nfi<br>rm<br>ed<br>!=<br>"Y<br>" |  | ni<br>sh<br>ed<br>=<br>Fa<br>ls<br>e |  |  |  | ut<br>pu<br>t                                                                                          |
|---|---|---|-----------------------|---|---|--|----------------|--|--|--|--|----------------------------------------|--|--------------------------------------|--|--|--|--------------------------------------------------------------------------------------------------------|
| 1 | 0 |   |                       |   |   |  |                |  |  |  |  |                                        |  |                                      |  |  |  |                                                                                                        |
| 2 |   | " |                       |   |   |  |                |  |  |  |  |                                        |  |                                      |  |  |  |                                                                                                        |
| 3 |   |   | N                     |   |   |  |                |  |  |  |  |                                        |  |                                      |  |  |  |                                                                                                        |
| 4 |   |   |                       | 5 |   |  |                |  |  |  |  |                                        |  |                                      |  |  |  |                                                                                                        |
| 5 |   |   |                       |   | 0 |  |                |  |  |  |  |                                        |  |                                      |  |  |  |                                                                                                        |
| 6 |   |   |                       |   |   |  |                |  |  |  |  |                                        |  |                                      |  |  |  | " W<br>el<br>co<br>me<br>to M<br>at<br>h<br>Q<br>ue<br>st!<br>"                                        |
| 7 |   |   |                       |   |   |  |                |  |  |  |  |                                        |  |                                      |  |  |  | "A<br>ns<br>w<br>er<br>a<br>se<br>rie<br>s<br>of<br>qu<br>es<br>tio<br>ns<br>to<br>sa<br>ve<br>th<br>e |





|    |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                                                                     |
|----|---|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "                                                                                                                                                                   |
| 12 | " |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                                                                     |
| 17 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                                                                     |
| 18 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                                                                     |
| 19 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "                                                                                                                                                                   |
| 20 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "L<br>oa<br>di<br>ng<br>"                                                                                                                                           |
| 21 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "                                                                                                                                                                   |
| 22 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                                                                     |
| 23 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                                                                                                                                                     |
| 24 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | "S<br>ho<br>wi<br>ng<br>pr<br>ev<br>io<br>us<br>sc<br>or<br>es<br>. If<br>nil<br>, th<br>er<br>e<br>ar<br>e<br>no<br>pr<br>ev<br>io<br>us<br>sc<br>or<br>es<br>. St |

|    |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                                                                      |
|----|--|----------|--|--|--|--|--|--|--|--|------------------|--|--|--|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  | ati<br>sti<br>cs<br>wil<br>l<br>on<br>ly<br>be<br>re<br>co<br>rd<br>ed<br>if<br>th<br>e<br>en<br>tir<br>e<br>ru<br>n<br>is<br>fin<br>is<br>he<br>d." |
| 25 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                                                                      |
| 26 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                                                                      |
| 27 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                                                                      |
| 28 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                                                                      |
| 29 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                                                                      |
| 30 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  | "                                                                                                                                                    |
| 31 |  |          |  |  |  |  |  |  |  |  | T<br>R<br>U<br>E |  |  |  |  |  |  |                                                                                                                                                      |
| 32 |  | Wi<br>ll |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |                                                                                                                                                      |
| 33 |  |          |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  | "C<br>on<br>fir<br>m<br>us                                                                                                                           |

[illegible]

[illegible]

|    |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |               |                                                                                                               |
|----|--|--|--|--|--|--|--|---------------|--|--|--|--|--|--|--|--|---------------|---------------------------------------------------------------------------------------------------------------|
|    |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |               | yo<br>u<br>ha<br>ve<br>a<br>str<br>ea<br>k<br>of<br>2,<br>yo<br>u<br>go<br>t<br>an<br>ex<br>tra<br>lif<br>e!" |
| 52 |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |               | "Y<br>ou<br>no<br>w<br>ha<br>ve<br>4<br>liv<br>es<br>"                                                        |
| 53 |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |               |                                                                                                               |
| 54 |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  | FA<br>LS<br>E |                                                                                                               |
| 57 |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |               |                                                                                                               |
| 58 |  |  |  |  |  |  |  | FA<br>LS<br>E |  |  |  |  |  |  |  |  |               |                                                                                                               |
| 59 |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |               |                                                                                                               |
| 60 |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |               | "L<br>oa<br>di<br>ng<br>B<br>O                                                                                |

|    |  |  |  |  |  |  |                       |  |  |  |  |        |        |  |  |  |  |                                                                                                                                                                                                                                                                                                                                 |
|----|--|--|--|--|--|--|-----------------------|--|--|--|--|--------|--------|--|--|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |  |  |  |  |  |  |                       |  |  |  |  |        |        |  |  |  |  | S<br>S<br>F<br>I<br>G<br>H<br>T”                                                                                                                                                                                                                                                                                                |
| 61 |  |  |  |  |  |  |                       |  |  |  |  |        |        |  |  |  |  | “H<br>e<br>l<br>i<br>e<br>s<br>t<br>h<br>e<br>M<br>a<br>t<br>h<br>I<br>n<br>q<br>u<br>i<br>s<br>i<br>t<br>o<br>r<br>,<br>t<br>h<br>e<br>f<br>i<br>n<br>a<br>l<br>b<br>o<br>s<br>s<br>.<br>B<br>e<br>w<br>a<br>r<br>e<br>o<br>f<br>h<br>i<br>s<br>d<br>e<br>a<br>d<br>l<br>y<br>q<br>u<br>e<br>s<br>t<br>i<br>o<br>n<br>s<br>..” |
| 62 |  |  |  |  |  |  | F<br>A<br>L<br>S<br>E |  |  |  |  |        |        |  |  |  |  |                                                                                                                                                                                                                                                                                                                                 |
| 63 |  |  |  |  |  |  |                       |  |  |  |  | T<br>R | T<br>R |  |  |  |  |                                                                                                                                                                                                                                                                                                                                 |

|    |  |  |  |   |   |  |          |               |  |          |  |  |               |               |   |  |  |                                                                |
|----|--|--|--|---|---|--|----------|---------------|--|----------|--|--|---------------|---------------|---|--|--|----------------------------------------------------------------|
|    |  |  |  |   |   |  |          |               |  |          |  |  | U<br>E        | U<br>E        |   |  |  |                                                                |
| 64 |  |  |  | 0 | 0 |  |          | FA<br>LS<br>E |  |          |  |  |               |               |   |  |  |                                                                |
| 65 |  |  |  |   |   |  |          |               |  |          |  |  |               |               |   |  |  | "S<br>tat<br>s!"                                               |
| 66 |  |  |  |   |   |  |          |               |  |          |  |  |               |               |   |  |  | "Y<br>ou<br>r<br>cu<br>rre<br>nt<br>str<br>ea<br>k<br>is<br>0" |
| 67 |  |  |  |   |   |  |          |               |  |          |  |  |               |               |   |  |  | "Li<br>ve<br>s<br>re<br>m<br>ai<br>ni<br>ng<br>:<br>0"         |
| 68 |  |  |  |   |   |  |          |               |  | TR<br>UE |  |  |               |               |   |  |  |                                                                |
| 69 |  |  |  |   |   |  | TR<br>UE |               |  |          |  |  |               |               |   |  |  |                                                                |
| 63 |  |  |  |   |   |  |          |               |  |          |  |  | FA<br>LS<br>E | FA<br>LS<br>E |   |  |  |                                                                |
| 70 |  |  |  |   |   |  |          |               |  |          |  |  |               |               |   |  |  |                                                                |
| 71 |  |  |  |   |   |  |          |               |  |          |  |  |               |               | T |  |  |                                                                |



|    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |             |  |  |  |
|----|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-------------|--|--|--|
|    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | R<br>U<br>E |  |  |  |
| 72 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |             |  |  |  |
| 73 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |             |  |  |  |

## Data Types

### Char and String

The char data type is a single character, and a string datatype is a sequence of characters. In the program, characters and strings are used to store letters and numbers alike. It is used in a variety of variables, such as 'username'. 'username' gets input from the user, and uses that data to create a unique username. This username can be stored so the users can easily see the username and related information about it.

### Floating Point

The floating point data type is a family of data types which allows programmers to store fractional and numerical decimal values. In my code, it is used to store non-integer values. Since the int data type only allows whole numbers to be stored, float opted for a better option to store data, as it can store both whole and fractional numbers. In my code, 'correct\_answer' is a floating point variable that stores float data, and it is compared to 'answer' to see if the user is correct.

### Boolean

Boolean is a binary variable which has two variables. It is commonly used for True/False statements. This is implemented in my program to act as conditions for certain parts of the algorithm. For example, the 'calculator' variable is a boolean, and tells the program whether or not it can be used by the user, depending on its value.

### Date and Time

In the program, a timer is used. By utilising the time module built into Python, a timer starts as soon as the questions begin, and stops when all the questions have been completed. This allows the user to compare times with their peers, to see who is the fastest.

# Data Dictionary

| Variables      | Data Type      | Format for Display | Size in Bytes | Size for Display | Description                                                     | Example  | Validation                                                               |
|----------------|----------------|--------------------|---------------|------------------|-----------------------------------------------------------------|----------|--------------------------------------------------------------------------|
| username       | string         | XXXX               | 1-256         | 1-256            | A user created ID that identifies each individual user          | W1lliam  | Must be made of characters which can be in a string                      |
| forward        | boolean        | XXXX               | 4-5           | 4-5              | Will the algorithm continue? True or False                      | True     | Must be either True or False. If not True, the program will not continue |
| lives          | integer        | N                  | 1             | 1                | Number which indicates how many lives the user has left         | 0        | Must be an integer $\geq 0$                                              |
| streak         | integer        | N                  | 1             | 1                | Number which indicates the current answer streak the user is on | 2        | Must be an integer $\geq 0$                                              |
| correct_answer | floating point | NNNNN.N<br>NN      | 8             | 8                | Number which stores the correct answer to a question            | 21102.73 | Must be a float                                                          |
| answer         | floating point | NNNNN.N<br>N       | 8             | 8                | User inputted float which acts as the answer, is compared to    | 21102.73 | Must be a float                                                          |

|                    |         |        |     |     |                                                                |        |                                                                          |
|--------------------|---------|--------|-----|-----|----------------------------------------------------------------|--------|--------------------------------------------------------------------------|
|                    |         |        |     |     | correct_answer to determine if the user is right               |        |                                                                          |
| calculator         | boolean | XXXX   | 4-5 | 4-5 | Does the user have a calculator? True or False                 | False  | Must be either True or False                                             |
| choice             | boolean | X      | 1   | 1   | What do you choose to do? 'A' to answer, 'C' to use calculator | A      | Must be either A or C                                                    |
| begin              | string  | X      | 1   | 1   | Waits for the user to press ENTER before continuing            | ENTER  | If not Enter, the program will not continue                              |
| username_confirmed | boolean | X      | 1   | 1   | Confirm Username? Y or N                                       | Y      | Must be Y or N                                                           |
| questions_finished | boolean | XXXX   | 4-5 | 4-5 | Are questions finished? True or False                          | True   | Must be True or False                                                    |
| continueo<br>n     | boolean | XXXX   | 4-5 | 4-5 | Will the algorithm continue? True or False                     | True   | Must be either True or False. If not True, the program will not continue |
| start              | time    | NN.NNs | 4   | 4   | Starts a timer, counting in seconds                            | 0.1s   | Must be counted in seconds                                               |
| end                | time    | NN.NNs | 4   | 4   | Stops the timer                                                | 95.65s | Must be counted in seconds                                               |

|                 |                |        |    |    |                                                     |                         |                         |
|-----------------|----------------|--------|----|----|-----------------------------------------------------|-------------------------|-------------------------|
| endtime         | floating point | NNN.NN | 5  | 5  | Calculates the time in minutes                      | 2.6                     | Must be a float and > 0 |
| statisticsArray | array          | XXXXXX | 5  | 5  | Array which stores username, streak, lives, endtime | ['William', 7, 4, 4.56] | Must be valid data      |
| f               | file object    | XXXXXX | 20 | 20 | Puts the statistics array into a sequential file    | ['William', 7, 4, 4.56] | Must be valid data      |

## Data Structures

### Arrays

Arrays are data structures which act as a container which can store many items of the same data type. They store data in an ordered manner which is specified by the algorithm. In my program, an array is used to store the user's data/statistics, allowing users to see their data after the program has been run. It works in tandem with a sequential file to do this.

### Sequential Files

Sequential files are data structures that hold records, allowing them to be accessed sequentially by the algorithm. Each record added is added to the end of the file. This is used in my program to store the user's statistics. Using the array, data is written to the sequential file. If there is no file, it will create one and input that data into it. The sequential file can be saved elsewhere, making it a perfect solution to save data to, improving the user experience.